

EXPERIMENT NO: 8

TITLE: Power Factor Improvement

AIM:

To improve the power factor of a three-phase load and determine the reduction in line current after power factor correction.

PROBLEM STATEMENT:

A three-phase load operates at:

Power = 100 kW

Voltage = 400 V

Initial Power Factor = 0.65

Improved Power Factor = 0.90

Determine the line current before and after power factor correction and calculate the percentage reduction in current.

GIVEN DATA:

Power, $P = 100 \text{ kW} = 100000 \text{ W}$

Voltage, $V = 400 \text{ V}$

Initial Power Factor = 0.65

Improved Power Factor = 0.90

FORMULA:

Current,

$$I = P / (\sqrt{3} \times V \times \text{PF})$$

Percentage Reduction,

$$\% \text{ Reduction} = [(I_1 - I_2) / I_1] \times 100$$

CALCULATIONS:

1. Current Before Power Factor Correction

$$I_1 = 100000 / (1.732 \times 400 \times 0.65)$$

$$I_1 = 222 \text{ A}$$

2. Current After Power Factor Correction

$$I_2 = 100000 / (1.732 \times 400 \times 0.90)$$

$$I_2 = 160 \text{ A}$$

3. Percentage Reduction

% Reduction

$$= [(222 - 160) / 222] \times 100$$

$$= 27.93 \%$$

SOLUTION:

The current before power factor correction is found to be 222 A.

The current after power factor correction is found to be 160 A.

The reduction in current due to improvement in power factor is 27.93%.

RESULT:

Current Before Correction = 222 A

Current After Correction = 160 A

Percentage Reduction = 27.93 %

Hence the power factor of the system is improved successfully

EXPERIMENT NO: 10

TITLE: Weighted Least Square (WLS) State Estimation

AIM:

To estimate the state variables of a power system using the Weighted Least Square (WLS) method.

PROBLEM STATEMENT:

Given the measurement vector and measurement matrix, estimate the system state variables using the Weighted Least Square (WLS) technique.

GIVEN DATA:

Measurement Vector:

$$z = [1.02, 0.98, 1.03]$$

Measurement Matrix:

$$H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

Weight Matrix:

$W = \text{Identity Matrix}$

FORMULA:

State Estimate:

$$x = (H^T W H)^{-1} H^T W z$$

Since $W = I$,

$$x = (H^T H)^{-1} H^T z$$

CALCULATIONS:

Step 1:

$$H^T H = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Step 2:

$$(H^TH)^{-1} =$$

$$(1/3)$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Step 3:

$$H^Tz =$$

$$\begin{bmatrix} 2.05 \\ 2.01 \end{bmatrix}$$

Step 4:

Estimated State Vector

$$x =$$

$$\begin{bmatrix} 0.7725 \\ 0.7325 \end{bmatrix}$$

SOLUTION:

The state variables are estimated using the WLS algorithm.

The estimated values obtained are:

$$x_1 = 0.7725$$

$$x_2 = 0.7325$$

RESULT:

Estimated State Vector:

$$x = [0.7725 \ 0.7325]$$

Hence the state variables are successfully estimated using the WLS method.

EXPERIMENT NO: 11

TITLE: SCADA and RTU Simulation

AIM:

To simulate the operation of a SCADA system and RTU for monitoring electrical parameters.

PROBLEM STATEMENT:

A SCADA system measures a voltage of 230 V and a current of 10 A. Determine the monitored power.

GIVEN DATA:

Voltage (V) = 230 V

Current (I) = 10 A

FORMULA:

Power,

$$P = V \times I$$

CALCULATIONS:

$$P = 230 \times 10$$

$$P = 2300 \text{ W}$$

SOLUTION:

The power monitored by the SCADA system is obtained by multiplying the measured voltage and current.

$$P = 2300 \text{ W}$$

RESULT:

Power Monitored = 2300 W

Hence the SCADA system successfully monitors the electrical power.

EXPERIMENT NO: 12

TITLE: Bad Data Detection in State Estimation

AIM:

To detect erroneous measurements in a power system using residual analysis.

PROBLEM STATEMENT:

Given a set of measurement residuals and a specified threshold, identify the bad measurements.

GIVEN DATA:

Residual Vector:

$$r = [0.006, 0.492, -0.492, 0.496]$$

Threshold Value = 0.1

FORMULA:

If

$$|\text{Residual}| > \text{Threshold}$$

Then the measurement is considered bad data.

CALCULATIONS:

Measurement 1:

$$|0.006| < 0.1$$

Good Data

Measurement 2:

$$|0.492| > 0.1$$

Bad Data

Measurement 3:

$$|-0.492| > 0.1$$

Bad Data

Measurement 4:

$$|0.496| > 0.1$$

Bad Data

SOLUTION:

The residual values of measurements 2, 3 and 4 exceed the specified threshold value.

Therefore these measurements are identified as bad data.

RESULT:

Bad Measurements = 2, 3 and 4

Bad Data Vector = $[0 \ 1 \ 1 \ 1]$

Hence the bad data are successfully detected.

EXPERIMENT NO: 13

TITLE: Economic Dispatch of Thermal Generating Units

AIM:

To determine the optimal generation schedule of thermal generating units for minimum operating cost while satisfying the load demand.

PROBLEM STATEMENT:

A power system has a total load demand of 250 MW. Determine the optimal power generation schedule and operating cost.

GIVEN DATA:

Total Load Demand (PD) = 250 MW

Generator 1 Output (P1) = 160 MW

Generator 2 Output (P2) = 90 MW

Operating Cost = Rs. 2394.01/hour

FORMULA:

Total Generated Power,

$$P_T = P_1 + P_2$$

Power Balance Condition,

$$P_T = P_D$$

CALCULATIONS:

Step 1:

Generator 1 Output

$$P_1 = 160 \text{ MW}$$

Step 2:

Generator 2 Output

$$P_2 = 90 \text{ MW}$$

Step 3:

Total Generated Power

$$P_T = P_1 + P_2$$

$$P_T = 160 + 90$$

$$P_T = 250 \text{ MW}$$

Step 4:

Check Power Balance

$$P_T = P_D$$

$$250 \text{ MW} = 250 \text{ MW}$$

Hence the load demand is satisfied.

Step 5:

Operating Cost

$$\text{Cost} = \text{Rs. } 2394.01/\text{hour}$$

SOLUTION:

The economic dispatch problem allocates the generation among the available units such that the operating cost is minimum.

Generator 1 generates 160 MW and Generator 2 generates 90 MW.

The total generated power is equal to the load demand of 250 MW.

The minimum operating cost obtained is Rs. 2394.01/hour.

RESULT:

Generator 1 Output = 160 MW

Generator 2 Output = 90 MW

Total Generated Power = 250 MW

Minimum Operating Cost = Rs. 2394.01/hour

Hence the economic dispatch is successfully carried out.

EXPERIMENT NO: 14

TITLE: Unit Commitment

AIM:

To determine the optimal combination of generating units that satisfies the load demand at minimum operating cost.

PROBLEM STATEMENT:

Three generating units with capacities of 100 MW, 150 MW and 200 MW are available. Determine the optimal unit commitment schedule for a load demand of 150 MW.

GIVEN DATA:

Generating Unit Capacities:

Unit 1 = 100 MW

Unit 2 = 150 MW

Unit 3 = 200 MW

Load Demand = 150 MW

Optimal Schedule = [1 1 0]

Operating Cost = Rs. 1200/hour

FORMULA:

Total Available Capacity

PT = Sum of committed units

CALCULATIONS:

Step 1:

Optimal Unit Schedule

[1 1 0]

Where,

1 = Unit ON

0 = Unit OFF

Therefore,

Unit 1 = ON

Unit 2 = ON

Unit 3 = OFF

Step 2:

Calculate Available Capacity

$$PT = 100 + 150$$

$$PT = 250 \text{ MW}$$

Step 3:

Operating Cost

$$\text{Cost} = \text{Rs.}1200/\text{hour}$$

SOLUTION:

The optimal commitment schedule is obtained as $[1 \ 1 \ 0]$.

This indicates that Units 1 and 2 are committed while Unit 3 remains off.

The available generation capacity is 250 MW, which is sufficient to meet the required load demand.

The minimum operating cost obtained is Rs.1200/hour.

RESULT:

Optimal Unit Commitment Schedule = $[1 \ 1 \ 0]$

Available Capacity = 250 MW

Minimum Operating Cost = Rs.1200/hour

Hence the optimal unit commitment schedule is obtained successfully.

EXPERIMENT NO: 15

TITLE: Demand Side Management (DSM)

AIM:

To study the effect of Demand Side Management techniques on reducing the peak load demand of a power system.

PROBLEM STATEMENT:

The peak load before DSM implementation is 450 MW and after DSM implementation is 420 MW. Determine the peak reduction and percentage reduction in peak demand.

GIVEN DATA:

Peak Load Before DSM = 450 MW

Peak Load After DSM = 420 MW

FORMULA:

Peak Reduction

= Peak Load Before DSM – Peak Load After DSM

Percentage Reduction

= (Peak Reduction / Peak Load Before DSM) $\times$ 100

CALCULATIONS:

Step 1:

Peak Reduction

= 450 – 420

= 30 MW

Step 2:

Percentage Reduction

= (30 / 450) $\times$ 100

= 6.67%

SOLUTION:

After implementing Demand Side Management techniques, the peak load is reduced from 450 MW to 420 MW.

The reduction in peak demand is 30 MW.

The percentage reduction achieved is 6.67%.

RESULT:

Peak Load Before DSM = 450 MW

Peak Load After DSM = 420 MW

Peak Reduction = 30 MW

Percentage Reduction = 6.67%

Hence the peak load demand is successfully reduced using Demand Side Management techniques.

.